

Investigation of the Environmental Risk Factors Associated with Abnormally Elevated Cancer Incidence on Staten Island, New York

Paul M. Khoury¹

¹Weill Cornell Medicine Summer Research Program, New York, NY

Corresponding Author: Dr. Christopher E. Mason

Weill Cornell Medicine, 1305 York Ave., Y13-05, New York, NY 10021

Abstract

Staten Island, New York City's least populous borough, has the highest incidence rate of invasive malignant tumors of the five boroughs, roughly 20% above the citywide average. This report reviews existing epidemiological literature and publicly available registry data to evaluate whether proximity to the borough's former landfills – Fresh Kills, Great Kills, and Brookfield Park – and the chemical and radiological waste they contain can account for this excess. Geographic, temporal, and dose-response analyses are considered alongside confounding socioeconomic and behavioral risk factors, including household income, smoking prevalence, and ambient air quality. The evidence assembled here is consistent with the hypothesis that long-term, low-level exposure to carcinogenic and radioactive waste contributes to Staten Island's elevated cancer burden, and cannot be explained by income, smoking, or air quality alone. Directions for confirmatory epidemiological and molecular studies are proposed.

Keywords: cancer epidemiology; Staten Island; Fresh Kills landfill; environmental carcinogens; thyroid cancer; radioactive waste

1. Background and Problem

Staten Island – New York City's least populous borough, sometimes referred to as “the Forgotten Borough” – has the highest incidence rate of invasive malignant tumors in the city, about 20% higher than the rest of the city¹. In fact, according to data from the New York State Department of Health, Staten Island has the highest incidence rates for most cancers¹ across sexes, compared to the rest of the city¹.

A widely suspected risk factor is the proximity of Staten Island residents to numerous former solid waste landfills and dumping grounds, such as Fresh Kills, Great Kills, and Brookfield Park. Fresh Kills, at its peak the largest landfill in the world, took in 29,000 tons of trash per day, including an assortment of carcinogenic industrial chemicals². Of particular interest to this report are the numerous shipments Fresh Kills received from Ground Zero of the September 11th terrorist attacks, from the day of the attacks through June of the following year³. The latter two landfills took in a comparatively smaller, but still massive, amount of waste – on the order of millions of cubic yards of garbage, including untreated radioactive waste, primarily from products containing radium common throughout the earlier half of the twentieth century⁴. A survey conducted by the National Park Service found 1,200 radioactive hot spots throughout present-day Great Kills Park⁵.

2. Objective and Hypothesis

The objective of this work is to elucidate which environmental risk factors play the most significant role in the anomalous incidence of cancers on Staten Island relative to the rest of the five boroughs, through a holistic review of existing literature and analysis of relevant data.

We hypothesize that Staten Islanders face an increased risk of developing cancer compared to other New York City residents due to long-term, low-level exposure to carcinogenic chemical and radioactive waste, which is found at hazardous levels in and around the borough's former landfills.

3. Methods and Results

A comprehensive literature search and subsequent analysis yielded the following findings.

3.1. Geographic proximity to landfills

A correlation exists between geographic proximity, within Staten Island, to the landfills and cancer incidence. Residents of Staten Island Community District 3^{1,6} – which includes Tottenville, Great Kills, Eltingville, and Annadale, and which contains two former landfills and borders a third – have an 8.4% higher incidence rate of cancer than their counterparts on the North Shore.

3.2. Thyroid cancer and the 9/11 connection

In the 17 years immediately following September 11, 2001, and the consequent relocation of Ground Zero rubble to Fresh Kills, incidence of thyroid cancer on Staten Island increased

¹Including Hodgkin's lymphoma, non-Hodgkin's lymphoma, leukemia, and cancers of the esophagus, colorectum, pancreas, lung, urinary bladder, kidney and renal pelvis, brain and nervous system, and thyroid.

by approximately 300%⁷. The onset of this rapid increase corresponds with the arrival of toxic, smoldering rubble from Ground Zero at Fresh Kills (Fig. 1).

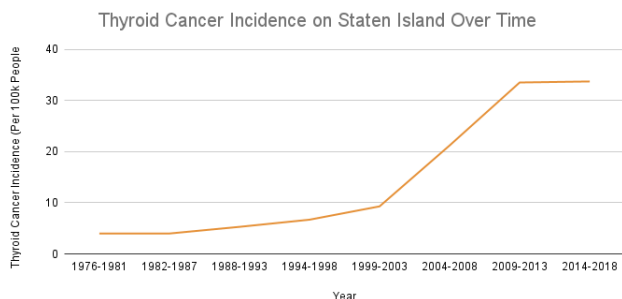


Figure 1: Thyroid cancer incidence on Staten Island over time, in cases per 100,000 people. Data retrieved from the New York State Department of Health Cancer Registry⁷.

Among 9/11 first responders, incidence of specific cancers was similarly elevated. For thyroid cancer in particular, incidence was elevated by up to 239% compared with a control population^{8,9}. This mirrors the trend observed on Staten Island, where thyroid cancer incidence among South Shore/Community District 3 residents is 140% higher than that of the city as a whole¹.

Some studies have attributed the markedly elevated incidence of thyroid cancer among 9/11 first responders and survivors to increased monitoring and screening rather than a true excess. However, Tuminello et al. conducted a study that refuted a monitoring and screening bias for thyroid cancer⁸.

Cancer Incidence (Per 100k People) vs. Median Household Income

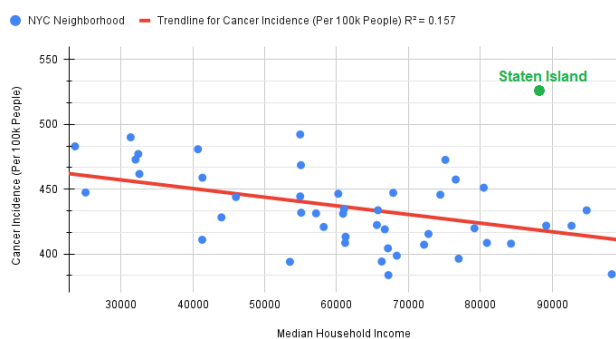


Figure 2: Cancer incidence (per 100,000 people) versus median household income across New York City neighborhoods. Staten Island (green) has both a high median household income and the highest cancer incidence, despite the negative citywide trend ($R^2 = 0.157$). For this analysis, Staten Island's three PUMA-designated community districts were treated as a single neighborhood. Data retrieved from the New York State Department of Health Cancer Registry¹.

3.3. Household income as a confounder

Despite every neighborhood in Staten Island having one of the highest median household incomes in the city – a characteristic typically correlated with *lower* cancer incidence – Staten Island and each of its constituent neighborhoods still have the highest incidence of cancer in the city¹⁰ (Fig. 2). This suggests that socioeconomic status does not explain the observed excess.

3.4. Smoking and air quality

The author also examined another common cancer risk factor: smoking. While Staten Island has the highest percentage of smokers of the five boroughs (just 0.2 percentage points above the Bronx), the incidence of cancer on Staten Island far exceeds the value predicted from smoking rates alone¹¹ (Fig. 3), pointing to the presence of additional risk factors. Furthermore, Staten Island has some of the best ambient air quality in the city, which rules out poor local air quality as a driving factor¹².

Smoking Population vs. Cancer Incidence by Borough

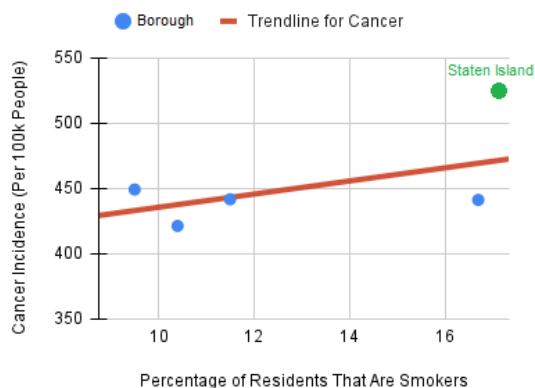


Figure 3: Smoking prevalence versus cancer incidence by New York City borough. Staten Island (green) falls well above the citywide trendline, indicating that cancer incidence exceeds what smoking prevalence alone would predict. Data retrieved from NYC Smoke-Free, a program of Public Health Solutions¹¹.

4. Future Directions

Considerably more research is required to establish, with high confidence, the existence of a cancer cluster on Staten Island, and whether it arises from radioactive waste, chemical waste, waste from the 9/11 response, or some combination thereof. We propose two large-scale public health investigations to extend this work. The first would collect detailed medical and behavioral information from consenting New York City residents to determine whether behavioral risk factors are disproportionately elevated among Staten Island residents in a way that could contribute to the borough's excess cancer in-

cidence. The second would be a genealogical survey to quantify DNA double-strand breaks (DSBs) among Staten Island residents. DSBs are a biomarker of radiation- and chemical-induced DNA and cellular damage and are mechanistically linked to carcinogenesis. A positive correlation between an individual's proximity to a former landfill and their measured DSB burden would corroborate the hypothesis that Staten Island's landfills are toxic to nearby residents and may be elevating their risk of developing cancer. Taken together, these findings underscore the need for coordinated environmental and epidemiological monitoring on Staten Island, particularly given the borough's unique history of waste disposal and its lingering effects on public health.

Beyond further monitoring, these findings support concrete intervention. The New York City Department of Health and Mental Hygiene, in coordination with the NYS Department of Environmental Conservation, should expand subsidized thyroid and cancer screening programs specifically for residents of Community District 3, prioritizing those with the longest residential history near Fresh Kills, Great Kills, and Brookfield Park. Remediation efforts at these sites should be reassessed and, where soil or groundwater testing indicates ongoing contamination, accelerated rather than treated as complete. Finally, the city should establish a public, regularly updated registry tracking cancer incidence by neighborhood alongside landfill remediation status, giving residents and clinicians the data needed to make informed health and housing decisions while the underlying research continues.

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